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Total No. of Pages : 03

Total No. of Questions : 09

MCA (2015 & Onwards) (Sem.-5)
ARTIFICIAL INTELLIGENCE
Subject Code : MCA-501
Paper ID : [74381]

Time : 3 Hrs.

Max. Marks : 60

INSTRUCTIONS TO CANDIDATES :

1. **SECTIONS-A, B, C & D** contains **TWO** questions each carrying **TEN** marks each and students has to attempt any **ONE** question from each **SECTION**.
2. **SECTION-E** is **COMPULSORY** consisting of **TEN** questions carrying **TWENTY** marks in all.
3. Use of non-programmable **scientific calculator** is allowed.

SECTION-A

1. (a) You have three jugs measuring 12 litres, 8 litres, & 3 litres, and a water faucet.

You need to measure out exactly one litre. Use state space approach to solve this problem.

(b) What is a Turing test and what capabilities are required by a computer to pass this test? Explain
2. (a) Explain the basic procedure for solving a problem in AI.

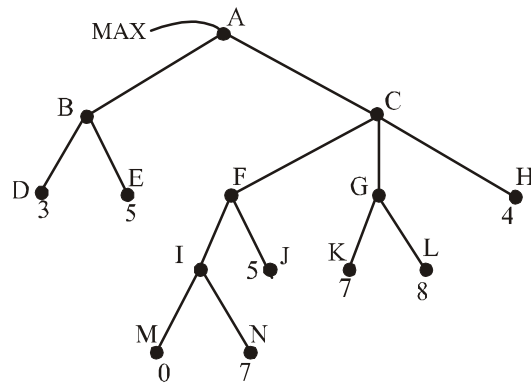
(b) How searching is used to provide solutions also describe some real world problems?

SECTION-B

3. (a) How does iterative deepening help? For search in binary tree, show four iterations using iterative deepening.

(b) Differentiate tree based breadth-first and depth-first search strategies based on completeness, time and space complexities.

4. Consider the following game tree.



- Find the best move for the MAX player using the minimax procedure
- Perform a left-to-right alpha-beta pruning on the tree. Indicate where the cutoffs occur.

SECTION-C

5. (a) Consider the statements in the following paragraph:

“Every human, animal and bird is living thing who breathe and eat. All birds can fly. All man and woman are humans who have two legs. Cat is an animal and has a fur. All animals have skin and can move. Giraffe is an animal who is tall and has long legs. Parrot is a bird and is green in color”

For the above paragraph, provide predicate logic representations.

- Describe various knowledge representation issues.
6. Explain the resolution algorithm for propositional logic. Now consider the following set of facts as the knowledge :
- The humidity is high and the sky is cloudy.
 - If the sky is cloudy then it will rain.
 - If the humidity is high it is hot.
 - It is not hot.

SECTION-D

7. (a) Discuss types of reasoning in Bayesian Networks. Explain briefly how Bayesian statistics provides reasoning under various kinds of uncertainty.

- (b) Convert the following sentence into conceptual dependency:

I heard a lion in the jungle

8. (a) What are the common techniques used for parsing of natural language structures?

Explain with example.

- (b) What is Fuzzy Logic? Explain with the help of examples, how is it different from the crisp logic?

SECTION-E

9. **Write briefly :**

- (a) Define minimax value with regards to optimal decision in games.
- (b) List merits and demerits of frames in Knowledge Representation.
- (c) Write various applications of AI.
- (d) Explain the term Modus Ponens with example.
- (e) State the significance of using heuristic function.
- (f) What is the difference between informed & uninformed search techniques?
- (g) What do you mean by uncertainty? Why uncertainty arises?
- (h) What do you understand by reasoning?
- (i) Define AI techniques.
- (j) Differentiate between syntax and semantics, of languages.